Kmeans聚类算法实现

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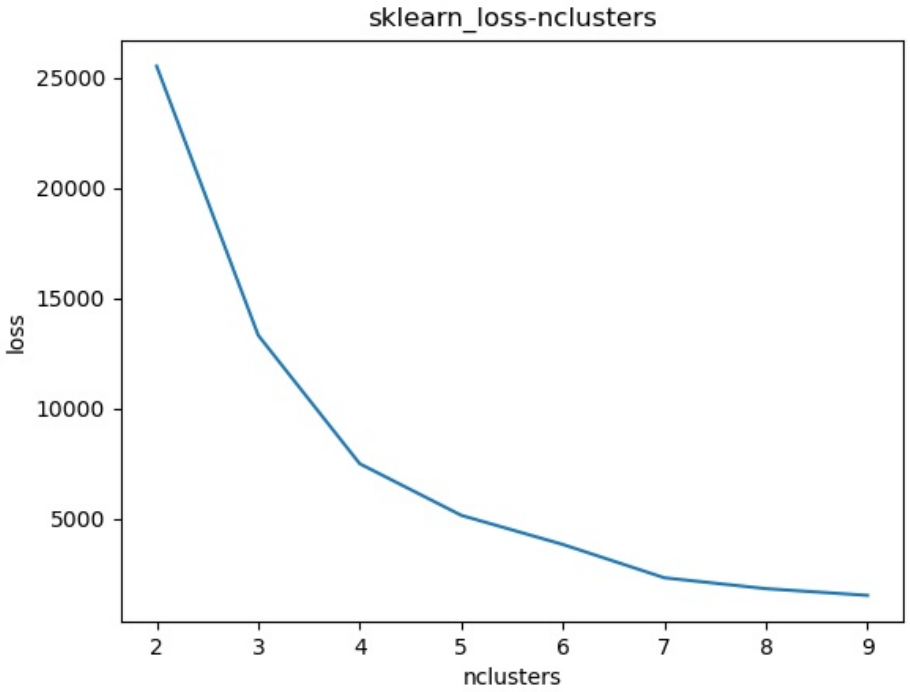
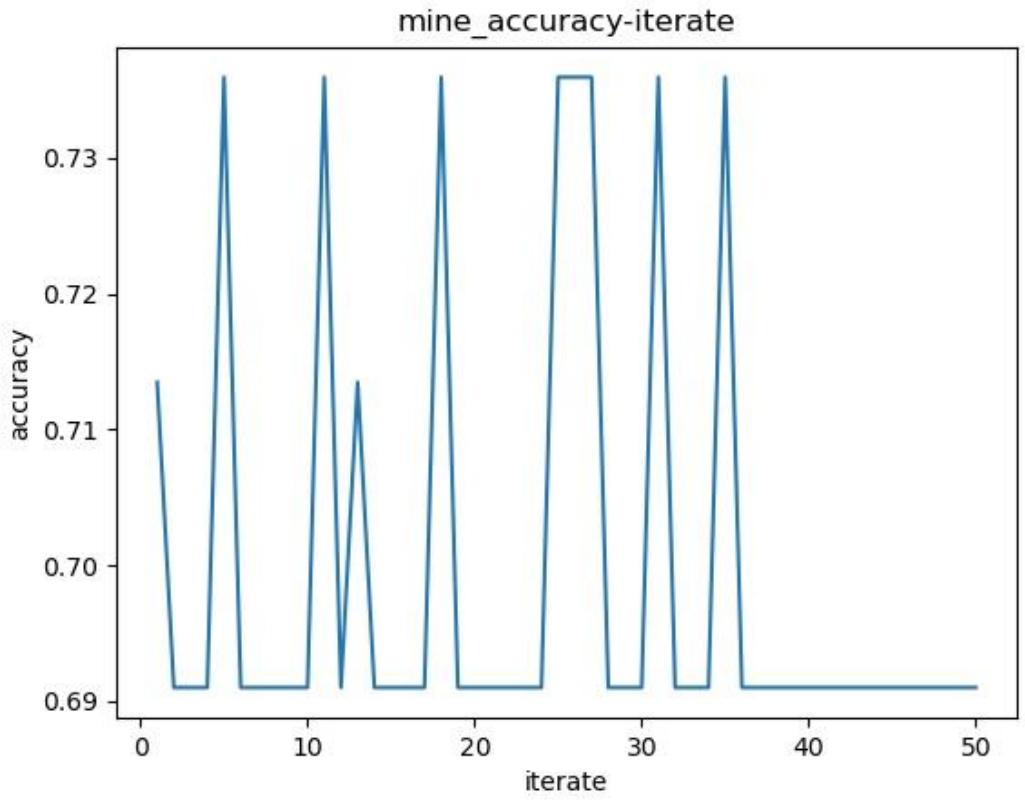
2020.4.2

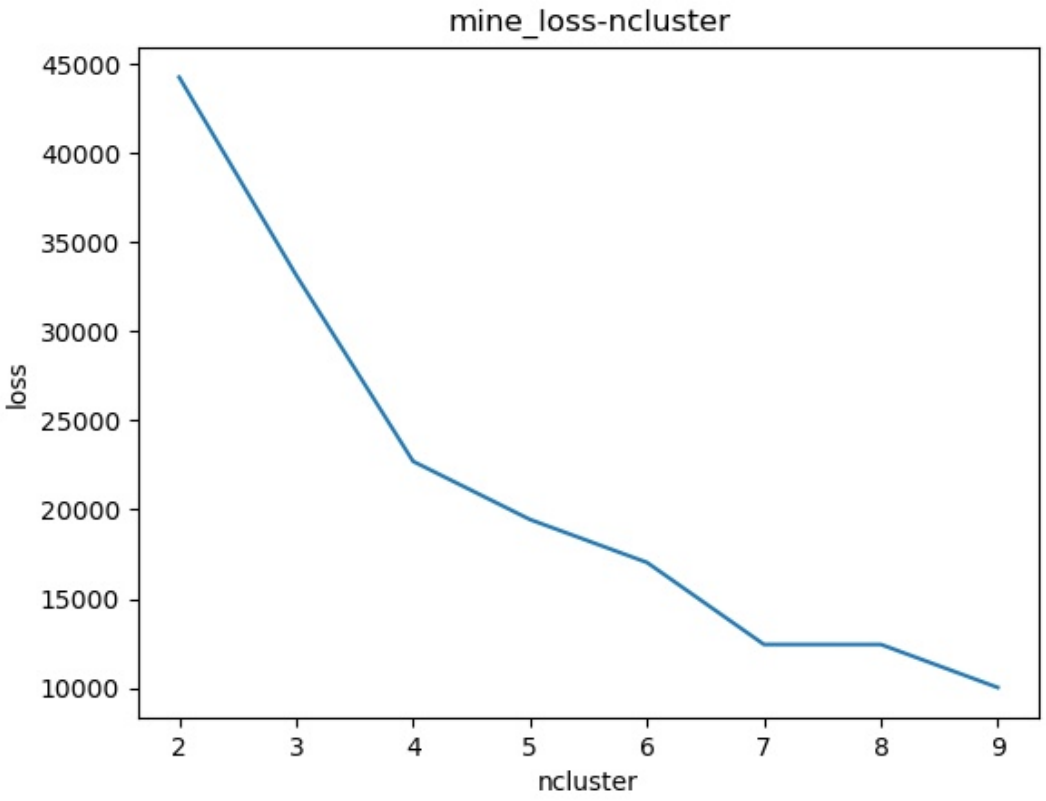
sklearn_kmeans

```
data.shape,label.shape (178, 13) (178,)
预测标签 (178,)
[1 1 1 1 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 2 2 2 1 1 2 2 1 1 2 1 1 1 1 1 1 2 2
 1 1 2 2 1 1 2 2 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 2 0 2 0 0 2 0 0 2 2 2 0 0 1
 2 0 0 0 2 0 0 2 2 0 0 0 0 0 2 2 0 0 0 0 0 2 2 0 2 0 2 0 0 0 2 0 0 0 0 2 0
 0 2 0 0 0 0 0 0 0 2 0 0 0 0 0 0 0 0 0 2 0 0 2 2 2 2 0 0 0 2 2 0 0 2 2 0 2
 2 0 0 0 0 2 2 2 0 2 2 2 0 2 0 2 2 0 2 2 2 2 0 0 2 2 2 2 2 0]
err(178): 43 err1(59): 14 err2(71): 26 err3(48): 3
accuracy: 0.7584269662921348
```

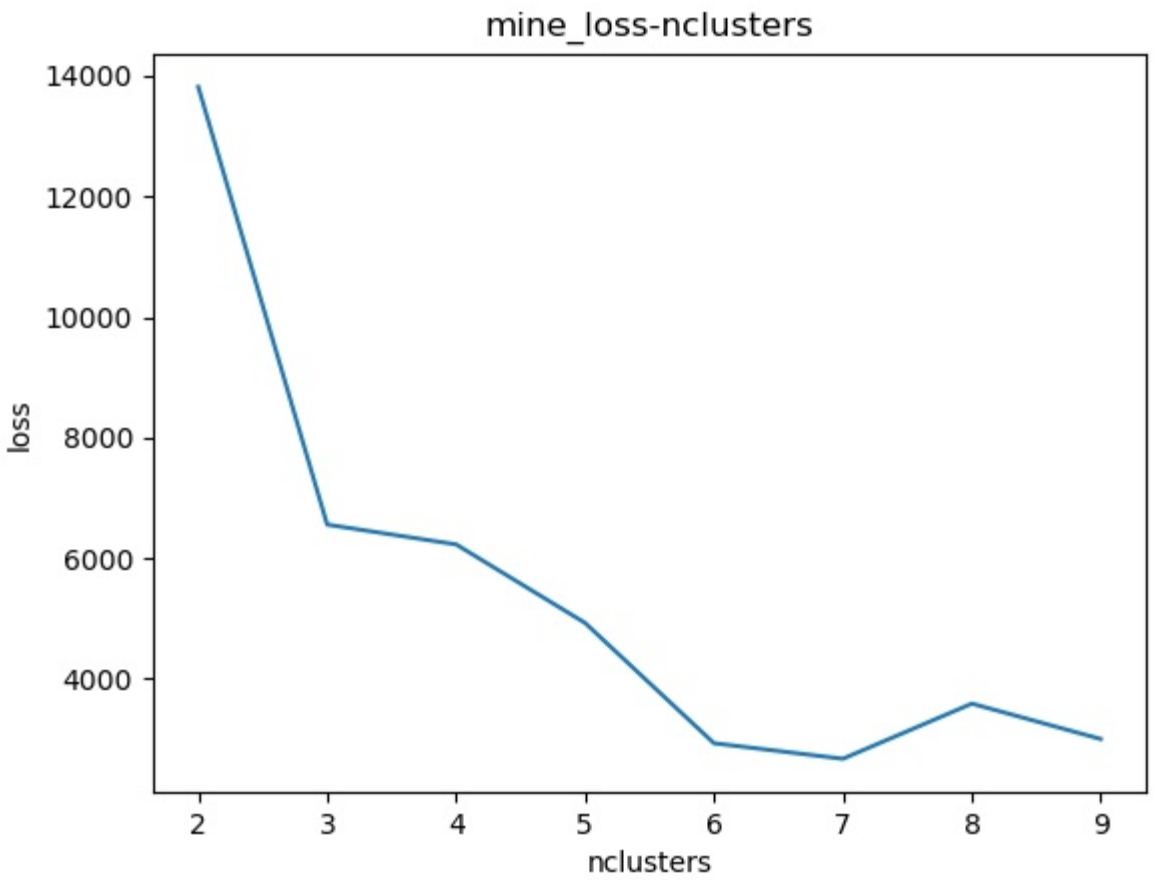


mine_kmeans


mine_accuracy-iterate
accuracy
0.73
0.72
0.71
0.70
0.69
0
10
20
30
40
50
iterate


mine_loss-ncluster
loss
45000
40000
35000
30000
25000
20000
15000
10000
2
3
4
5
6
7
8
9
ncluster

mine_kmeans



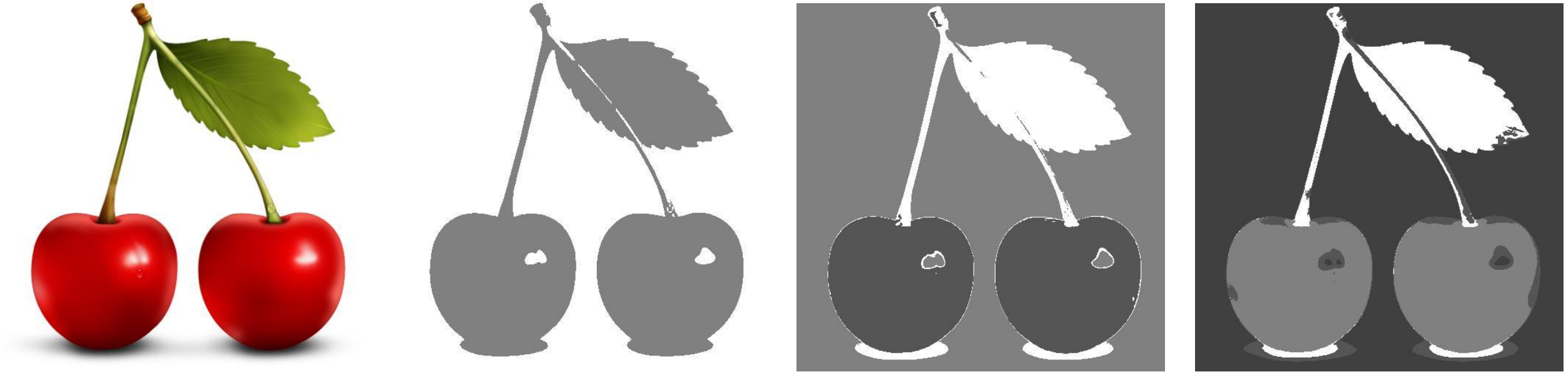
3 0 0.6910112359550562

3 7 0.7134831460674158

3 12 0.6910112359550562

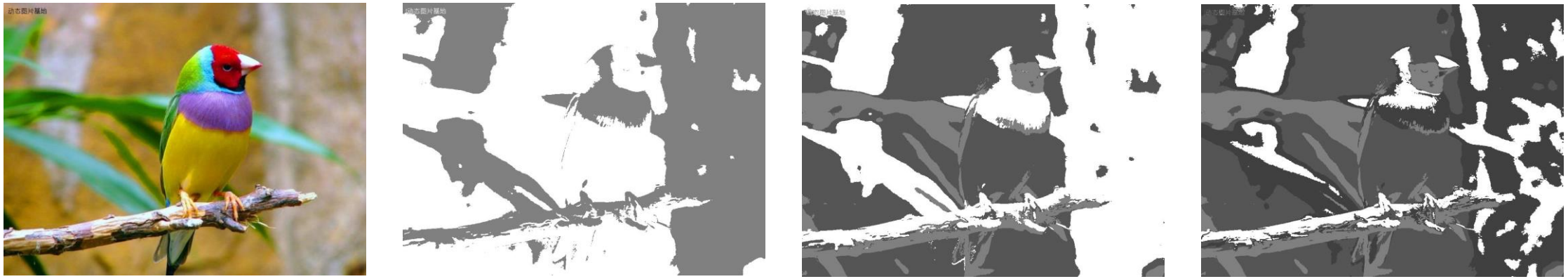
3 35 0.6910112359550562

mine_kmeans



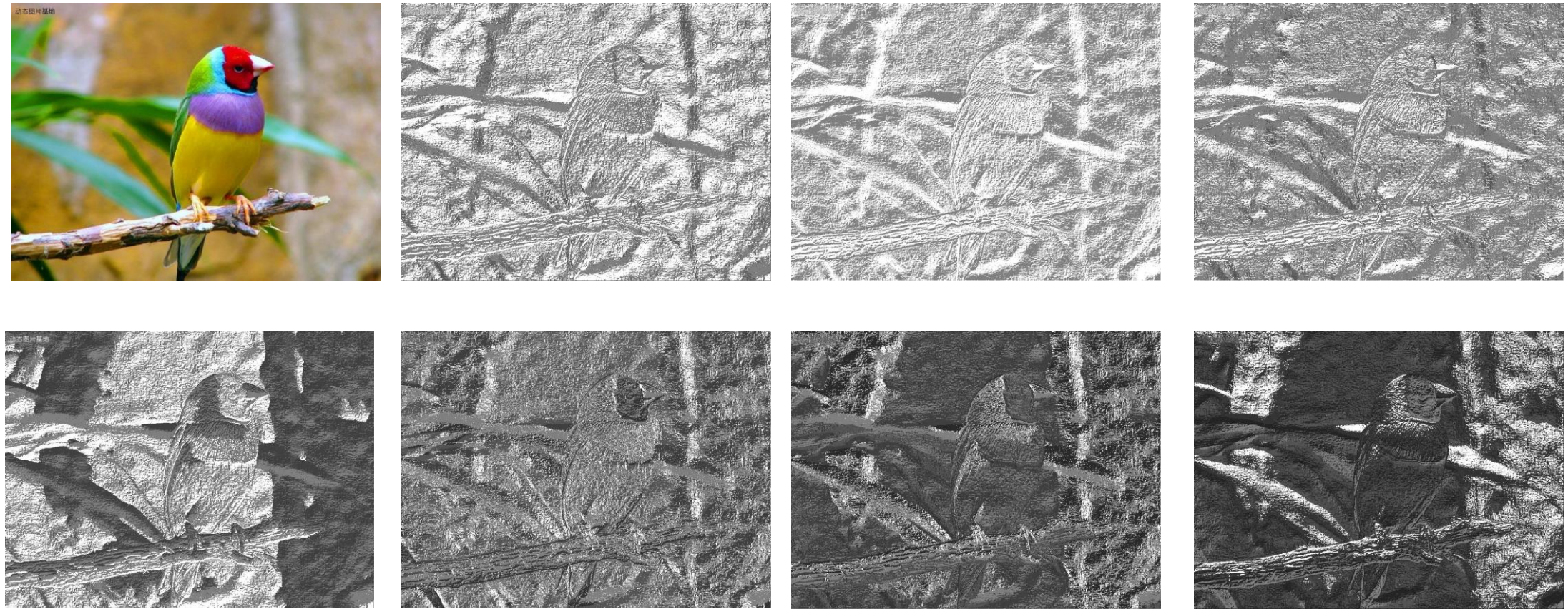
RGB+直接聚类2-4类

mine_kmeans



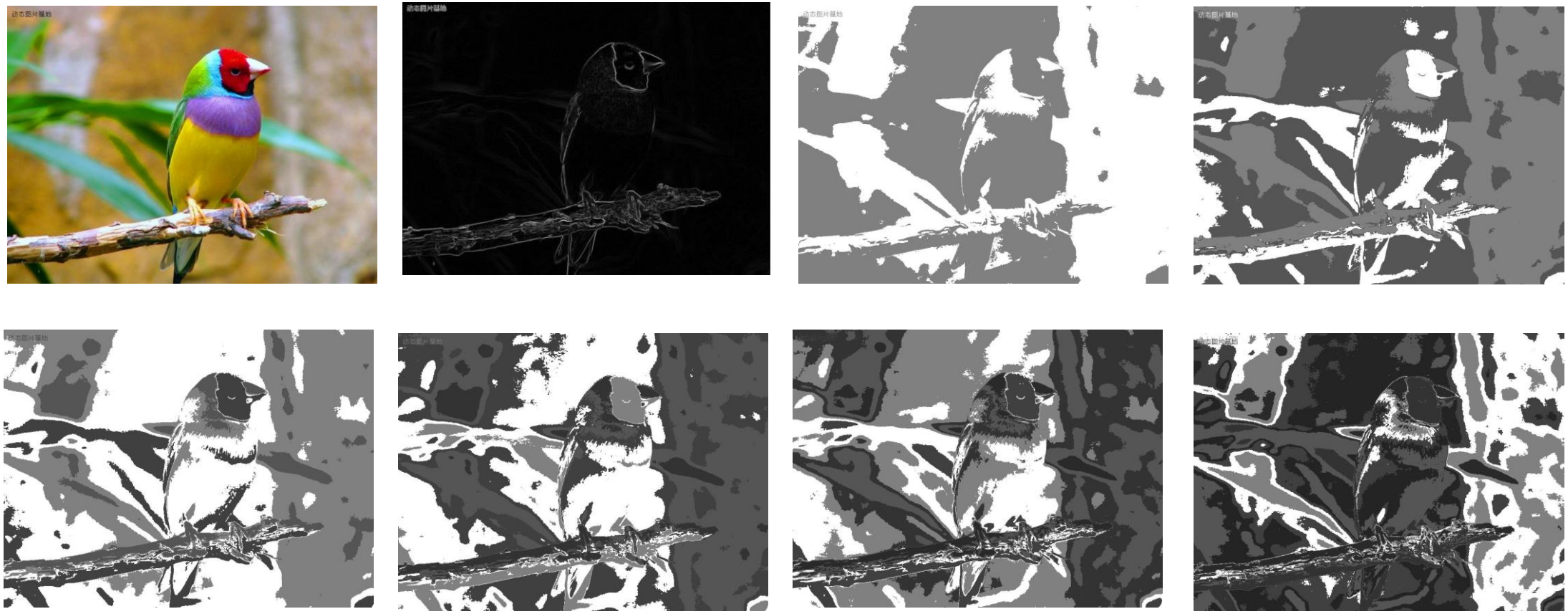
RGB+直接聚类2-4类

LBP+mine_kmeans



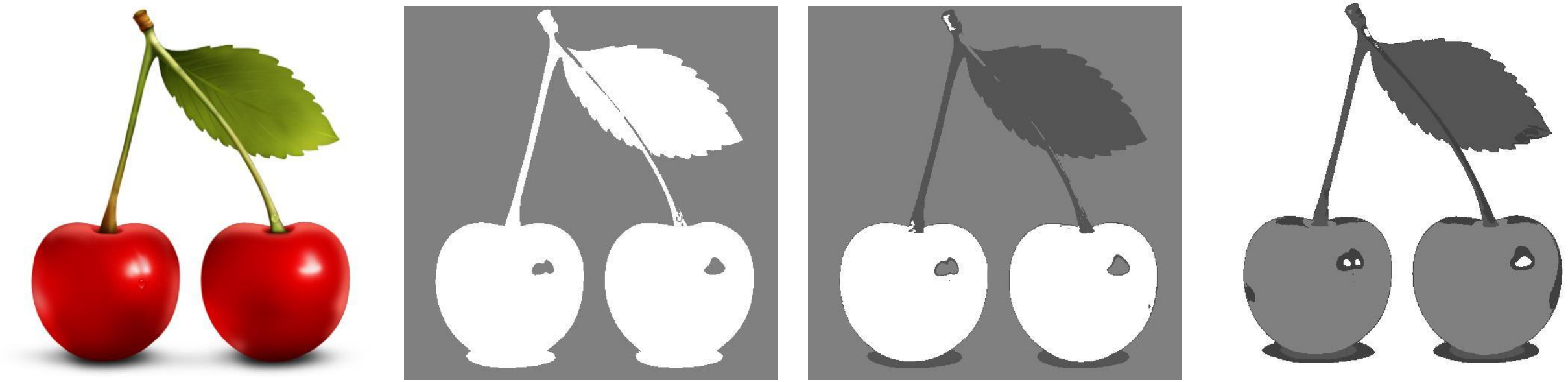
RGB+直接提取纹理+纹理_聚类2-7类

Edge+mine_kmeans



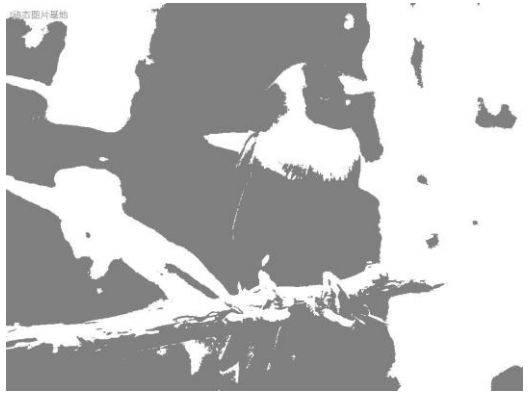
RGB+直接提取边缘+边缘_聚类2-7类

sklearn_kmeans



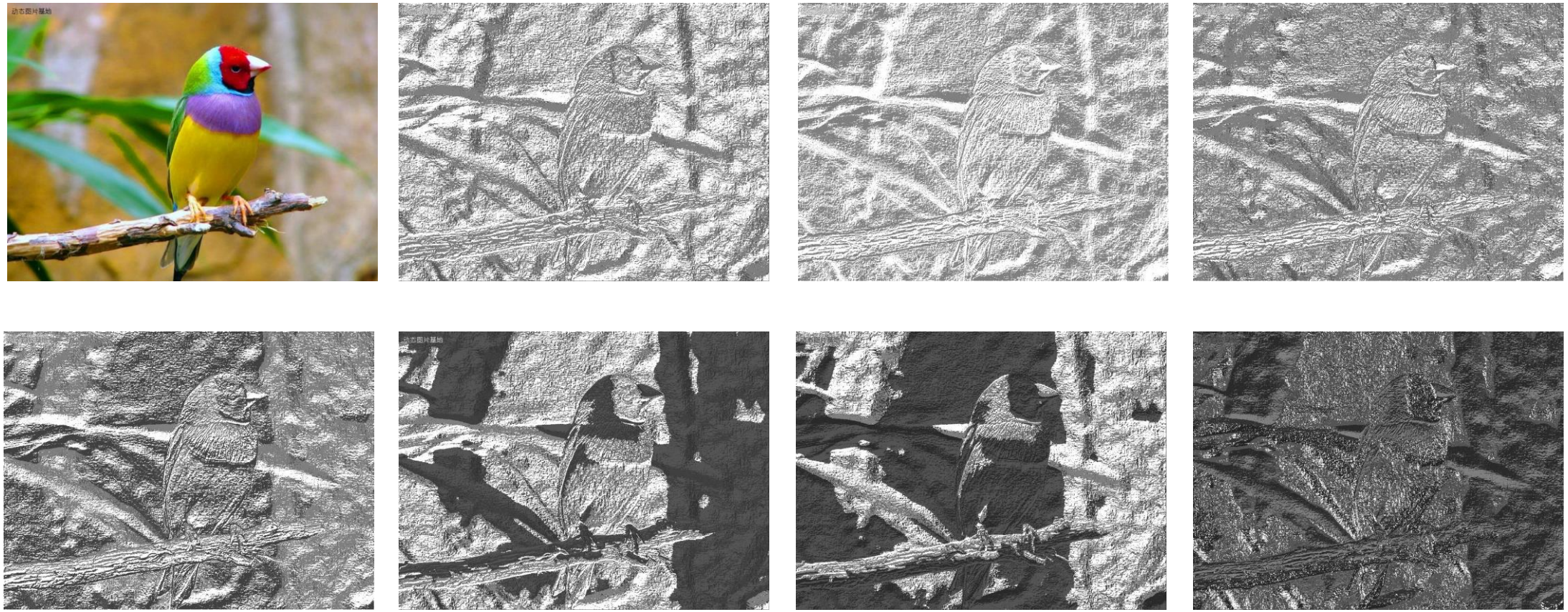
RGB+直接聚类2-4类

sklearn_kmeans



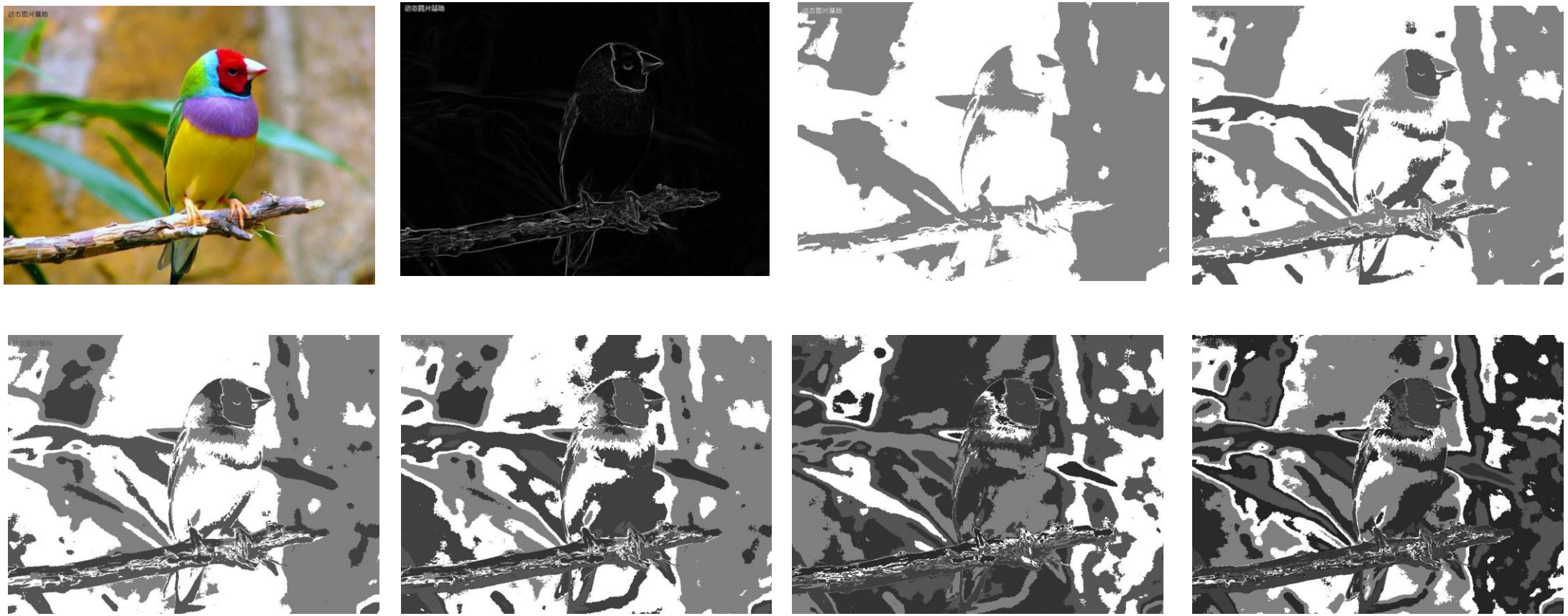
RGB+直接聚类2-4类

LBP+sklearn_kmeans



RGB+直接提取纹理+纹理_聚类2-7类

Edge+sklearn_kmeans



RGB+直接提取边缘+边缘_聚类2-7类

总结

- 改变初始类心，聚类准确度发生改变。
- 改变聚类类别数，loss（平均类内距离平方和）发生改变，分类数越多，loss越小。

Thank You!